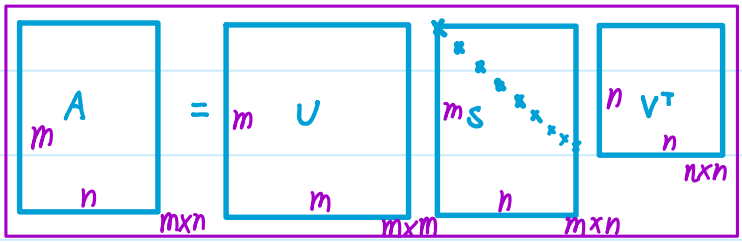
Week 15, Tuesday

Quiz for 30 minutes

△ Singular value decomposition (SVD)

Def: $\forall\ A \in \mathbb{R}^{m\times n}$ with $m \geq n$, $A = USV^T$, where $U \in \mathbb{R}^{m\times m}$, $S \in \mathbb{R}^{m\times n}$, $V \in \mathbb{R}^{n\times n}$, and U, V are orthogonal and S is given in the form: $S = \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_n \end{bmatrix}$

$\sigma_i \geq 0$: singular value.



$A = USV^T$.

$U^TU = I_{m\times m}$.

$V^TV = I_{n\times n}$.

Rk: ① $AV = US$

② $AA^T = USV^TVS^TU^T = USS^TU^T$, i.e.

$(AA^T)U = U\Lambda_1$, where $\Lambda_1 = \begin{bmatrix} \sigma_1^2 & & & & & \\ & \sigma_2^2 & & & & \\ & & \ddots & & & \\ & & & \sigma_n^2 & & \\ & & & & 0 & \\ & & & & & \ddots \\ & & & & & & 0 \end{bmatrix} \in \mathbb{R}^{m\times m}$

$AA^T\vec{u}_j = \sigma_j^2\vec{u}_j$

$\therefore (\vec{u}_j, \sigma_j^2)$ is an eigenpair of AA^T

③ Similarly: $A^TAV = V\Lambda_2$, where $\Lambda_2 = \begin{bmatrix} \sigma_1^2 & & & \\ & \sigma_2^2 & & \\ & & \ddots & \\ & & & \sigma_n^2 \end{bmatrix} \in \mathbb{R}^{n\times n}$

$A^TA\vec{v}_j = \sigma_j^2\vec{v}_j$

$\therefore (\vec{v}_j, \sigma_j^2)$ is an eigenpair of A^TA

$\boxed{\text{Thm}}$ $A \in \mathbb{R}^{m\times n}$

(i) A^TA and AA^T are symmetric and positive semidifinite

(ii) $N(A) = N(A^TA)$

(iii) $\text{rank}(A) = \text{rank}(A^TA) = \#$ of nonzero singular values

(iv) nonzero eigenvalues of AA^T and AA^T are the same.

SVD: ① $A^TA \in \mathbb{R}^{n\times n}$, symmetric, it can be diagonalized by eigenpairs

$(\sigma_1^2, \vec{v}_1)$, $(\sigma_2^2, \vec{v}_2)$, $\cdots$, $(\sigma_n^2, \vec{v}_n)$, where

$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > \sigma_{r+1} = \cdots = \sigma_n = 0$ $\quad S = \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_n \\ & & & \end{bmatrix}$

② orthogonalization: $\vec{v}_1, \cdots, \vec{v}_n$, $V = [\vec{v}_1, \vec{v}_2, \cdots, \vec{v}_n]$, $VV^T = I_n$.

then: $A^TA = VDV^T$ (diagonalization)

V: may not unique. $\quad D = S^TS = \begin{bmatrix} \sigma_1^2 & & & \\ & \sigma_2^2 & & \\ & & \ddots & \\ & & & \sigma_n^2 \end{bmatrix} \in \mathbb{R}^{n\times n}$

③ get U by: $AV = US \Rightarrow$

$$A\vec{v}_i = \sigma_i \vec{u}_i \Rightarrow \vec{u}_i = \frac{1}{\sigma_i} A\vec{v}_i, \quad i = 1, 2, \cdots, r$$

Verify: $\vec{u}_1, \cdots, \vec{u}_r$ orthonormal.

$$\vec{u}_i^T \vec{u}_j = \frac{1}{\sigma_i \sigma_j} \vec{v}_i^T A^T A v_j = \frac{\sigma_j^2}{\sigma_i \sigma_j} \vec{v}_i^T \vec{v}_j = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

Q: how to get remaining $m-k$ columns of U?

A: free to choose $m-k$ vectors in $\mathbb{R}^m$ which linearly independent with $\vec{u}_1, \cdots, \vec{u}_r$, as, $\{\vec{u}_1, \vec{u}_2, \cdots, \vec{u}_r, \vec{w}_{r+1}, \cdots, \vec{w}_n\}$

+ orthogonalization (Gram-Schmidt process)

$\Rightarrow \{\vec{u}_1, \vec{u}_2, \cdots, \vec{u}_n\} \Rightarrow U = [\vec{u}_1, \cdots, \vec{u}_n]$ with $U^TU = I_m$.

① ② ③ $\Rightarrow$ $A = USV^T$ (verify)

$\Leftrightarrow AV = US$, i.e. $AV = [A\vec{v}_1, A\vec{v}_2, \cdots, A\vec{v}_r, A\vec{v}_{r+1}, \cdots, A\vec{v}_n]$

$$= [\sigma_1\vec{u}_1, \sigma_2\vec{u}_2, \cdots \sigma_r\vec{v}_r, \vec{0}, \cdots, \vec{0}]$$

$$= [\vec{u}_1, \vec{u}_2, \cdots, \vec{u}_r, \vec{u}_{r+1}, \cdots, \vec{u}_n] \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_n \\ & & & \end{bmatrix} = US$$

RK: $\vec{v}_{r+1}, \cdots, \vec{v}_n$ basis of $N(A^TA)$, also basis of $N(A)$.

Q: Can we start with getting U by eigen-decomposition of AA^T?

A: theoritically Yes, practically No. Since AA^T: $m\times m \gg n\times n$ if $m \gg n$.

e.g. $A=\begin{bmatrix}1&0&1\\0&1&0\\0&1&1\\0&1&0\\1&1&0\end{bmatrix}$, $A^TA=\begin{bmatrix}2&1&1\\1&4&1\\1&1&2\end{bmatrix}$ $\quad \lambda_1=5,\ \vec{v}_1^T=[1,2,1]^T$

$\lambda_2=2,\ \vec{v}_2^T=[1,-1,1]^T$

$\lambda_3=1,\ \vec{v}_3^T=[-1,0,1]^T$

$$S=\begin{bmatrix}\sqrt5&0&0\\0&\sqrt2&0\\0&0&1\\0&0&0\\0&0&0\end{bmatrix},\quad V=\begin{bmatrix}\frac{1}{\sqrt6}&\frac{1}{\sqrt3}&-\frac{1}{\sqrt2}\\\frac{2}{\sqrt6}&-\frac{1}{\sqrt3}&0\\\frac{1}{\sqrt6}&\frac{1}{\sqrt3}&\frac{1}{\sqrt2}\end{bmatrix}$$

$$\vec{u}_1=\frac{1}{\sigma_1}A\vec{v}_1=\frac{1}{\sqrt5}A\vec{v}_1=\left[\frac{2}{\sqrt{30}},\frac{2}{\sqrt{30}},\frac{3}{\sqrt{30}},\frac{2}{\sqrt{30}},\frac{3}{\sqrt{30}}\right]^T$$

$$\vec{u}_2=\frac{1}{\sigma_2}A\vec{v}_2=\frac{1}{\sqrt2}A\vec{v}_2=\left[\frac{2}{\sqrt6},-\frac{1}{\sqrt6},0,-\frac{1}{\sqrt6},0\right]^T$$

$$\vec{u}_3=\frac{1}{\sigma_3}A\vec{v}_3=A\vec{v}_3=\left[0,0,\frac{1}{\sqrt2},0,-\frac{1}{\sqrt2}\right]^T$$

$$\vec{u}_4=\frac{1}{\sqrt6}[1,1,-1,1,-1]^T$$

$$\vec{u}_5=\frac{1}{\sqrt2}[0,1,0,-1,0]^T$$

$$U=[\vec{u}_1,\vec{u}_2,\vec{u}_3,\vec{u}_4,\vec{u}_5]=\begin{bmatrix}\frac{2}{\sqrt{30}}&\frac{2}{\sqrt6}&0&\frac{1}{\sqrt6}&0\\\frac{2}{\sqrt{30}}&-\frac{1}{\sqrt6}&0&\frac{1}{\sqrt6}&\frac{1}{\sqrt2}\\\frac{3}{\sqrt{30}}&0&\frac{1}{\sqrt2}&-\frac{1}{\sqrt6}&0\\\frac{2}{\sqrt{30}}&-\frac{1}{\sqrt6}&0&\frac{1}{\sqrt6}&-\frac{1}{\sqrt2}\\\frac{3}{\sqrt{30}}&0&-\frac{1}{\sqrt2}&-\frac{1}{\sqrt6}&0\end{bmatrix}\qquad A=USV^T$$

Applications: I: data compressing.

$$A=[\vec{u}_1,\vec{u}_2,\cdots,\vec{u}_r,\times,\cdots,\times]\begin{bmatrix}\sigma_1&&&&&\\&\sigma_2&&&&\\&&\ddots&&&\\&&&\sigma_r&&\\&&&&0&\\&&&&&\ddots\\&&&&&&0\end{bmatrix}\begin{bmatrix}\vec{v}_1^T\\\vec{v}_2^T\\\vdots\\\vec{v}_n^T\end{bmatrix}$$

$$=\sigma_1\vec{u}_1\vec{v}_1^T+\sigma_2\vec{u}_2\vec{v}_2^T+\cdots+\sigma_r\vec{u}_r\vec{v}_r^T$$

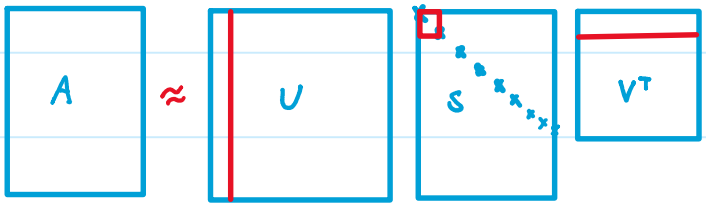
storage: $r(m+n)+r \ll mn$ if $r \ll \min\{m,n\}$

Rk: for full rank, $A=\sum_{i=1}^{n}\sigma_i\vec{u}_i\vec{v}_i^T$

if: $\sigma_1\sim\cdots\sim\sigma_r \gg \sigma_{r+1}\geq\cdots\geq\sigma_n$. or $|\sigma_{r+1}|,\cdots|\sigma_n|$ very close 0.

if: $\sigma_1 \sim \cdots \sim \sigma_r \gg \sigma_{r+1} \geq \cdots \geq \sigma_n$. or $|\sigma_{r+1}|, \cdots |\sigma_n|$ very close 0.

Take $\tilde{A} = \sum_{i=1}^{r} \sigma_i \vec{u}_i \vec{v}_i^T$ to approximate A.



II: Least square: $\min \frac{1}{2}\|A\vec{x} - \vec{b}\|_2$

$\Rightarrow$ normal equation: $A^TA\vec{x} = A^T\vec{b}$

if $A = USV^T$, $A^TA\vec{x} = VS^TSV\vec{x} = VS^TU^T\vec{b}$

i.e. $DV^T\vec{x} = S^TU^T\vec{b}$, let: $\vec{z} = V^T\vec{x}$, $U^T\vec{b} = \vec{c}$

$\Rightarrow \sigma_i^2 z_i = \sigma_i c_i \Rightarrow z_i = \frac{c_i}{\sigma_i}$ $i = 1, 2, \cdots, r$

remaining $z_{r+1}, \cdots, z_n$ are free to choose.

usually prefer $z_{r+1} = \cdots = z_n = 0$.

principle component analysis

III PCA: reduction of dimensions

Given data: $\vec{x}_j \in R^d$, $j = 1, 2, \cdots, N$, $d \gg 1$.

$X = [\vec{x}_1 \ \vec{x}_2 \ \cdots \ \vec{x}_N] \in R^{d \times N}$, $X = \underset{d\times d}{U}S\underset{N\times N}{V^T}$

find $P \in R^{k \times d}$ $(k \ll d)$ s.t. $\vec{y}_j = P\vec{x}_j \in R^k$, $j = 1, 2 \cdots, N$.

PCA requires: YY^T is diagonal, $Y = PX$

$YY^T = PXX^TP^T = PUSS^TU^TP^T :=$ diagonal $k \times k$

let: $PU = I_{k \times d}$, i.e. ith row of P = ith column of U, $i = 1, 2, \cdots, k$.

Rk: take $\vec{y}_j = P\vec{x}_j$ to reduce the dimensionality of data.

HW15-1: Sec 9.6: 1, 2 a, c 10, 12, 13, 14, 15, 17, 19